Aruna Thangavel

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Professional Summary

AI / LLM Engineer (Early Career) with hands-on experience building end-to-end Generative AI applications using Python, FastAPI, and LLM frameworks. Skilled in developing RAG-based systems, semantic search pipelines, and backend AI services. Currently focused on Agentic AI workflows, multi-step reasoning systems, and integrating LLMs with tools and APIs. Strong foundation in machine learning, NLP, and software engineering practices, with experience building scalable, production-oriented solutions.

Skills

- **Programming & Backend**: Python, SQL, Flask, REST API
- **Generative AI & LLMs**: RAG(Retrieval-Augmented generation), prompt engineering, Embeddings & Vector Search (FAISS), LangChain (Chains, Tools, Agents - basic) , LLM APIs (OpenAI-style usage) , Semantic Search Systems .
- **ML & NLP:** Scikit-learn, Pandas, Numpy, NLP-Tokenization,TF-IDF,Text Classification
- **AI Concepts:** Transformers (Attention, Embeddings- conceptual understanding
- **Tools & Engineering:** Git, API design, Basic Cloud Awareness(AWS fundamentals)

Projects

★ **LLM powered Medical Chatbot with RAG pipeline**

Technologies: Python, Langchain, FAISS, FastAPI/Flask

- *Built an end*-to-end RAG system for answering medical queries from documents.
- Built a prototype AI chatbot to answer questions from text/PDF documents using LLMs
- Explored embeddings and vector search using FAISS for retrieving relevant information
- Implemented basic prompt engineering techniques to improve response quality

★ **Predicting the success of telemarketing campaigns for bank institution**

Technologies: Python, Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn

- Built a machine learning classification system and explored feature importance to improve prediction accuracy, demonstrating strong data preprocessing and model evaluation skills.
- Analyzed the effectiveness of direct marketing campaigns via phone calls using classification algorithms like random forest which improved prediction accuracy by 74%.
- Identified key factors such as call duration and previous campaign outcomes that led to a 15% increase in customer subscription.

★ **Statistical Analysis and Modelling of Employee Attrition for HR Decision-Making**

Technologies: Python, Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn

- Analyzed employee attrition in an organization using statistical methods and machine learning models achieving an 85% prediction accuracy.
- Identified key drivers of attrition, contributing to a 10% reduction in overall employee turnover.

- ★ **Grocery Store Application**

  Technologies: Python, Vue Js, Flask, Sql-Alchemy
  - Developed a full-stack grocery store application, integrating Vue.js for front-end and Python with Flask for back-end.
  - Designed an intuitive user-friendly interface to facilitate 20% faster checkout process.
- ★ **Ticket Show Application**

  Technologies: HTML, CSS, Python, Flask
  - Implemented a web-based ticket booking application for users to browse shows, choose venues, select seats, and purchase tickets.
  - Enhanced user experience thereby increasing ticket booking efficiency by 25%.

Certifications

- Microsoft Certified: **Azure AI Fundamentals**
- Microsoft Certified: **Azure Data Fundamentals**

Education

- **Diploma in Data Science and Programming** | IIT Madras | 2025
- **Master of Technology (M.Tech)** | PSG College of Technology | 2012
- **Bachelor of Technology (B.Tech)** |Arulmighu Kalasalingam College of Engineering| 2010

Publications

- ★ **A Survey on Ontology Evaluation Tools**| Published in *IEEE Conference Proceedings* | 2012
  - Conducted an extensive survey of ontology evaluation tools and the key metrics.
- ★ **Weather Forecasting Using Back Propagation Neural Network and Regression Analysis**| Published in *Conference on Research Trends in IT, Biometrics and Cyber Security (CRT-ITBCS 2012)* | 2012
  - Built a neural network-based model for weather forecasting, improving prediction accuracy through regression analysis and neural networks.

Work Experience

- ★ ***Tata Consultancy Services Limited - Systems Engineer| 2016 – 2021***
  - Designed, implemented and executed 100+ automation test scripts to ensure the quality and reliability of banking applications and reducing manual testing efforts by 40%.
  - Gained foundational experience in automation fueling my transition to data science.

- ★ ***Cognizant Technology Solutions – Programmer Analyst| 2012 – 2016***
  - Provided technical support for java-based web applications, troubleshooting code-related issues, optimizing system performance and reducing downtime by 25%.